

Advanced Econometrics: Exercise #8

September 22, 2025

Professor Paulo Parente

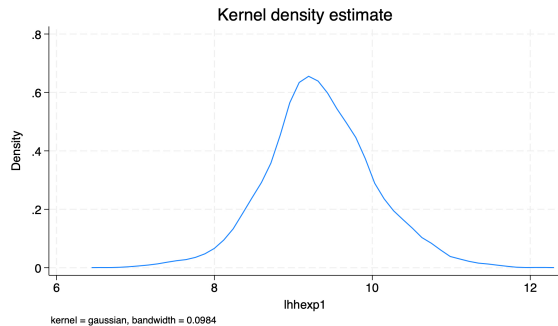
Amy Qian

Problem 1

Obtain a non-parametric estimate of the density of `lhhexp1`. Compare the bandwidth used by STATA with the one suggested by Silverman's rule of thumb.

STATA

```
kdensity lhhexp1, kernel(gaussian)
```



Bandwidth used by STATA: 0.0984

Silverman's Rule of Thumb

```
sum lhhexp1, d
```

This gives us

- Standard deviation: 0.6877458
- n: 5,999
- IQR: 9.759566 - 8.919547 = 0.840019

To calculate Silverman's bandwidth, we use the formula: $h = \min(\sigma, \frac{IQR}{1.34}) \frac{0.9}{n^{-1/5}}$

Plugging in our values, we get:

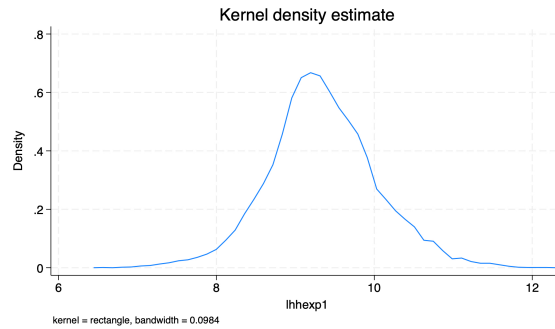
- `display ((.6877458*0.9)/5999^(0.2)) = .10865624`
- `display (((9.759566 - 8.919547)/1.34*0.9)/5999^(0.2)) = .09904009`

So the bandwidth suggested by Silverman's rule of thumb is 0.09904009, which is very close to the bandwidth used by STATA.

Problem 2

Repeat the previous exercise using different kernels.

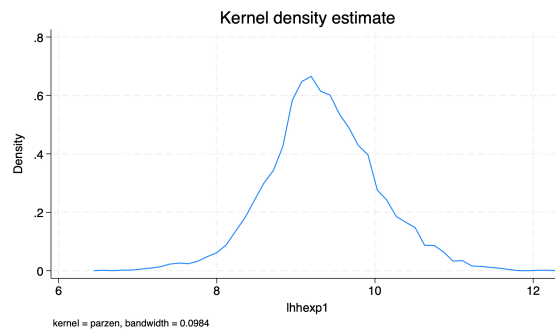
Rectangular Kernel



`kdensity lhexp1, kernel(rec)`

This is less smooth than the Gaussian kernel.

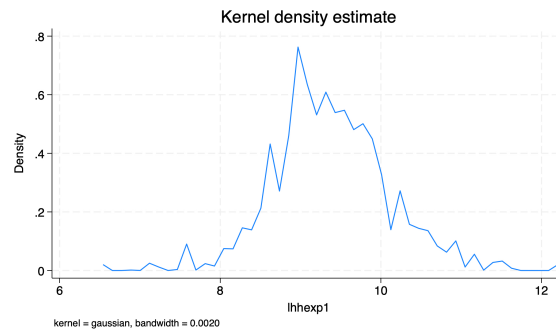
Parzan Kernel



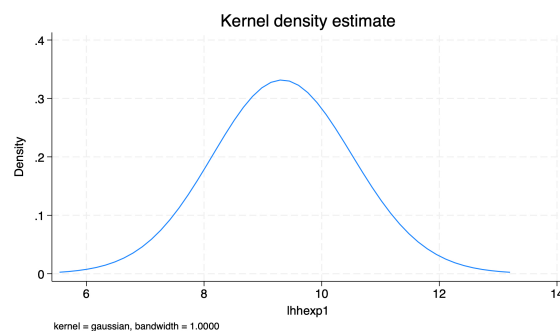
`kdensity lhexp1, kernel(parz)`

This is also less smooth than the Gaussian kernel.

Gaussian with different bandwidth



`kdensity lhexp1, kernel(gaussian) bwidth(0.002)`



`kdensity lhexp1, kernel(gaussian) bwidth(1)`

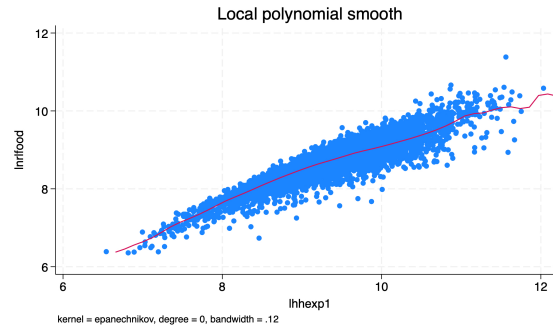
In general, the bandwidth is more important than the kernel. We want it to be smooth enough but not too smooth that we miss important features of the data.

Problem 3

Obtain the Nadaraya-Watson estimator of the conditional expectation of `lnrlfood`, the log household expenditure on food, given `lhhexp1`. Does the result suggest that this conditional expectation is linear?

Solution

We want to estimate $E[y_i|x_i] = m(x_i)$ where y_i is `lnrlfood`.



`lpoly lnrlfood lhhexp1`

Blue dots are the data points, the red line is the estimated conditional expectation (the function). Could be linear looking at the graph, but not exactly linear.

Problem 4

Using OLS, estimate a regression of `lnrlfood` on `lhhexp1`, `age` and its square. Is the effect of age linear?

Solution

First generate the new variable age squared.

`gen age2 = age^2`

Then run the OLS regression using robust standard errors to control for heteroskedasticity.

`reg lnrlfood lhhexp1 age age2, r`

We want to test the null hypothesis that the effect of age is linear, which is equivalent to testing if the coefficient of `age2` is zero.

$$H_0 : \beta_{\text{age2}} = 0 \text{ vs. } H_a : \beta_{\text{age2}} \neq 0$$

From the regression output, we see that the p-value is 0.000, which is less than 0.05, so we reject H_0 in favor of H_1 at 5% confidence level.

Problem 5

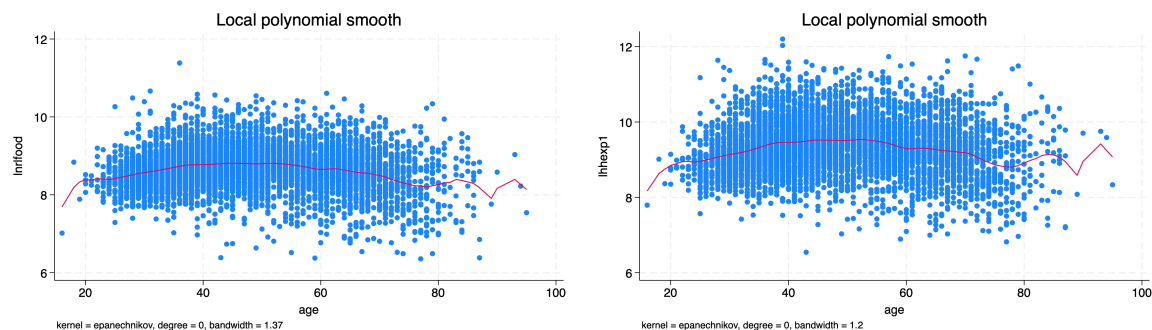
Use the partially linear estimator to estimate in the following model: $\text{lnrlfood}_i = \beta \text{lhhexp1}_i + g(\text{age}_i) + u_i$

Solution

$G(\cdot)$ is now an unknown function.

Get the estimates

- $E[y_i|\text{age}_i] \Rightarrow \text{foodage}$. `lpoly lnrlfood age, generate(foodage) at (age)`
- $E[x_i|\text{age}_i] \Rightarrow \text{expage}$. `lpoly lhhexp1 age, generate(expage) at (age)`



Now we compute the differences.

- $\text{gen } y = \text{lnrlfood} - \text{foodage}$
- $\text{gen } x = \text{lhhexp1} - \text{expage}$

Finally, we run the regression of y on x , not assuming anything on the distribution of the data.

```

Source |      SS      df      MS      Number of obs      =
> 5,999 |-----|-----|-----|-----|-----|-----|
> 64.09 |      1274.81681      1      1274.81681      Prob > F      =
> .0000 |      324.491644      5,998      .054099974      R-squared      =
> .7971 |-----|-----|-----|-----|-----|-----|
> .7971 |      1599.30845      5,999      .266595841      Adj R-squared      =
> 23259 |      1599.30845      5,999      .266595841      Root MSE      =

> -----|-----|-----|-----|-----|-----|
> y | Coefficient Std. err.      t      P>|t|      [95% conf. inte
> rval] |-----|-----|-----|-----|-----|-----|
> -----|-----|-----|-----|-----|-----|
> x |      .695743      .0045324      153.51      0.000      .686858      .70
> 46281 |-----|-----|-----|-----|-----|-----|
    
```

reg y x, noconstant

Problem 6

Compare the results.

```

Linear regression      Number of obs      =      5,999
                      F(3, 5995)          =      5484.44
                      Prob > F              =      0.0000
                      R-squared             =      0.8112
                      Root MSE            =      .2333
    
```

		Robust				
lnrlfood	Coefficient	std. err.	t	P> t	[95% conf. interval]	
lhhexp1	.6942858	.005711	121.57	0.000	.6830901	.7054815
age	.005095	.0016213	3.14	0.002	.0019168	.0082733
age2	-.0000716	.0000161	-4.45	0.000	-.0001031	-.00004
_cons	2.1277	.0561216	37.91	0.000	2.017681	2.237718

Actually quite close.